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James O. Jensen and David R. Yarkony

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On the evaluation of non-Born–Oppenheimer interactions for Born–Oppenheimer wave functions. V. A body fixed frame approach. Applications to isotope effects on equilibrium geometries and the adiabatic correction for the $X\,^1\Sigma^+$ state of LiH

James O. Jensen
Chemical Research, Development and Engineering Center, SMCCR-RSP-C, Aberdeen Proving Ground, Maryland 21010-5423

David R. Yarkony[a)]
Department of Chemistry, The Johns Hopkins University, Baltimore, Maryland 21218

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The evaluation of the total second derivative nonadiabatic coupling matrix element $H(J,I,\mathbf{R}) = \langle \psi_J(\mathbf{r};\mathbf{R}) | \Sigma_i (-1/2M_i)(\partial^2/\partial R_i^2) \psi_I(\mathbf{r};\mathbf{R}) \rangle_\mathbf{r}$ is considered. Here $\psi_J(\mathbf{r};\mathbf{R})$ is the adiabatic Born–Oppenheimer electronic wave function which in this work will be approximated by a large-scale CI wave function developed from an MCSCF reference space. For diatomic and triatomic systems the computational effort associated with the evaluation of $H(J,I,\mathbf{R})$ can be reduced considerably by the use of a body fixed frame approach. In this approach costly evaluation of the derivative wave function with respect to noninternal degrees of freedom in the space fixed frame is replaced by the evaluation of matrix elements of many electron operators including the mass polarization operator (total electronic linear momentum squared) and the L^2 operator (total electronic orbital angular momentum squared). The equivalence of the body fixed frame and space fixed frame results leads to valuable diagnostic equations which provide stringent tests of the derivative methodology used to evaluate the remaining second derivatives with respect to internal coordinates. The methods presented here are applied to the benchmark systems BeH^+ and LiH. The Born–Oppenheimer diagonal correction or adiabatic correction (AC) is evaluated for the $X\,^1\Sigma^+$ state of these systems and used to consider the effect of isotopic substitution on equilibrium geometries. For the $X\,^1\Sigma^+$ state of LiH a troubling discrepancy exists between the AC determined by advanced theoretical and experimental techniques. For $R \lesssim R_e$ the AC determined directly with specialized CI wave functions and the experimental value inferred from a detailed spectroscopic analysis of the $A \rightarrow X$ emission agree. However, for $R > R_e$ theory and experiment disagree qualitatively. For $R \lesssim R_e$ our results are consistent with the previous work. For $R > R_e$ our results are in accord with the experimentally derived AC.

I. INTRODUCTION

The development of theoretical methods to treat electronically nonadiabatic processes has long been a topic of considerable interest.[1–5] Recently it has been shown (parts I–IV of this series) that the analytic derivative techniques used in the evaluation of the derivatives of the Born–Oppenheimer energy $E_I(\mathbf{R})$ [$E_I(\mathbf{R}) = \langle \psi_I(\mathbf{r};\mathbf{R}) | \hat{H}^e \psi_I(\mathbf{r};\mathbf{R}) \rangle_\mathbf{r}$], can be used to facilitate the computation of the derivative coupling matrix elements required for the description of such processes in terms of Born–Oppenheimer adiabatic states. In particular, it was shown that using these techniques the derivative couplings

$$g_\alpha(J,I,\mathbf{R}) \equiv \left\langle \psi_J(\mathbf{r};\mathbf{R}) \left| \frac{\partial}{\partial R_\alpha} \psi_I(\mathbf{r};\mathbf{R}) \right\rangle_\mathbf{r} \right., \tag{1.1}$$

$$k_{\alpha,\beta}(J,I,\mathbf{R}) \equiv \left\langle \frac{\partial}{\partial R_\alpha} \psi_J(\mathbf{r};\mathbf{R}) \left| \frac{\partial}{\partial R_\beta} \psi_I(\mathbf{r};\mathbf{R}) \right\rangle_\mathbf{r} \right., \tag{1.2}$$

and

$$h_{\alpha,\beta}(J,I,\mathbf{R}) \equiv \left\langle \psi_J(\mathbf{r};\mathbf{R}) \left| \frac{\partial^2}{\partial R_\alpha \partial R_\beta} \psi_I(\mathbf{r};\mathbf{R}) \right\rangle_\mathbf{r} \right. \tag{1.3}$$

could be evaluated without approximation and without recourse to divided difference differentiation. In the above $\mathbf{r}$ and $\mathbf{R}$ are the space fixed frame coordinates of the electrons and nuclei, respectively, and $\hat{H}^e$ is the electronic Hamiltonian in the Born–Oppenheimer approximation. The Born–Oppenheimer electronic wave functions $\psi_J(\mathbf{r};\mathbf{R})$ are approximated by large-scale (10^5–10^6 terms) CI wave functions developed from MCSCF reference spaces and orbitals. The development of these techniques complements recent advances in the theoretical treatment of the dynamics of nonadiabatic processes,[2,4] and vibronic coupling in Jahn–Teller systems[10–12] as well as advances in experimental techniques in high resolution spectroscopy of diatomic[13–17] and triatomic[18–19] systems and state to state chemical dynamics.[20,21] As a result of these advanced experimental techniques increasingly accurate measurements of electronically nonadiabatic effects are now possible offering stringent tests of the theoretical methods.

In this work we consider further the efficient character-

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ization of nonadiabatic interactions between adiabatic states described by large-scale CI wave functions focusing on the adiabatic, or Born–Oppenheimer diagonal, correction, $H(I,I,\mathbf{R})$ to a potential energy surface (PES), $E_I(\mathbf{R})$ and the second derivative nonadiabatic coupling between electronic states $H(J,I,\mathbf{R})$ where (in atomic units)

$$H(J,I,\mathbf{R}) = \left\langle \psi_J(\mathbf{r};\mathbf{R}) \left| \sum_\alpha \frac{-1}{2M_\alpha} \frac{\partial^2}{\partial R_\alpha^2} \psi_I(\mathbf{r};\mathbf{R}) \right\rangle \right. \tag{1.4a}$$

$$= \sum_\alpha \frac{-1}{2M_\alpha} h_{\alpha,\alpha}(J,I,\mathbf{R}) \tag{1.4b}$$

and for $J = I$, $H(I,I,\mathbf{R}) = -K(I,I,\mathbf{R})$, where

$$K(J,I,\mathbf{R}) = \sum_\alpha \frac{-1}{2M_\alpha} k_{\alpha,\alpha}(J,I,\mathbf{R}). \tag{1.5}$$

In particular we will be concerned with efficiencies to be obtained from the ability to make maximal use of molecular point group symmetry in evaluating $H(J,I,\mathbf{R})$. This is accomplished by replacing the space fixed frame (SFF) evaluation of Eq. (1.4) by a body fixed frame (BFF)[2,5,22–25] procedure which decomposes Eq. (1.4) into a $\hat{P}^2$ or mass polarization contribution [originating in the transformation to the center-of-mass fixed frame (CMFF)], an $\hat{L}^2$ or angular momentum coupling contribution (originating in the transformation to the BFF), and a derivative $[h_{\alpha,\beta}(J,I,\mathbf{R})]$ contribution from *internal* coordinates only. Here $\hat{\mathbf{P}}$ is the total electronic linear momentum operator and $\hat{\mathbf{L}}$ is the total electronic orbital angular momentum operator. The precise form of these interactions is given in Sec. II.

Handy *et al.*[26] have observed that the derivative methodology noted above makes the evaluation of the adiabatic correction (AC) (Born–Oppenheimer diagonal correction), $-K(I,I,\mathbf{R})$, particularly simple in the SFF, that is directly from Eq. (1.4), especially if one or two configuration SCF wave functions are used. However, for diatomic and triatomic systems considerable computational efficiency is achieved through the use of a BFF procedure, especially when large-scale CI wave functions are used. These efficiencies will be shown to obtain from the ability to take full advantage of molecular point group symmetry.

Before proceeding with the development it is relevant to place this work in the context of previous theoretical treatments. The seminal work in this area is the entirely nonadiabatic treatment of H_2 in the BFF by Kolos and Wolniewicz.[23] Adiabatic states treatments of systems involving more than two electrons have emphasized the AC. The most extensive work in this area is that of Wolfsberg and coworkers[27–29] who used conventional SCF and small CI wave functions constructed from contracted Gaussian functions. These workers employed a BFF formalism, considered diatomic and triatomic systems and used their results to consider the effects of the AC on isotope exchange equilibria and spectral constants. This pioneering work was performed without the advantage of the derivative methodology noted above. More recently, Handy *et al*[26] evaluated the AC directly in the SFF using SCF and specialized two configuration SCF wave functions by taking advantage of derivative techniques and considered the effect of the AC on the singlet–triplet separation in methylene. Bishop and Cheung[30–32] evaluated the AC for the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states of LiH. These authors used an axis system attached to the geometric center of the molecule and specialized CI wave functions involving an elliptical orbital basis. Their adiabatically corrected potential energy curve (PEC) for the $X\,^1\Sigma^+$ state has been compared to that inferred by Stwalley, Vidal, and coworkers from highly resolved $A\,^1\Sigma^+ \rightarrow X\,^1\Sigma^+$ emission data.[15,16] Reasonable agreement for near equilibrium geometries was found. However, the agreement for larger internuclear separations is less satisfactory. Finally, Peric *et al.*[33] and Saxe and Yarkony[9] have considered particular contributions to the AC using generally applicable CI wave functions. The study of Saxe and Yarkony, which was performed in the vicinity of a symmetry allowed crossing of two PES's of the LiH_2 system, also considered the nonadiabatic interaction $H(J,I,\mathbf{R})$.

In this work we present the first study of the full AC using generally applicable CI wave functions and derivative techniques within a BFF formalism. The computational efficiencies inherent in this approach will be demonstrated. Using equivalences which relate SFF and BFF operators valuable relations will be derived which provide extremely stringent tests of the algorithms employed in the evaluation of the $\mathsf{k}(J,I,\mathbf{R})$. The $X\,^1\Sigma^+$ state of the benchmark systems BeH^+ and LiH will be considered. The BeH^+ system, which was considered by Kleinman and Wolfsberg[27] at the SCF level and Bishop and Cheung[34] using the specialized wave functions noted above, was chosen as a benchmark system since the large separation of the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states, $T_e = 39\,417\ \mathrm{cm}^{-1}$,[35] will minimize the importance of interstate nonadiabatic coupling. The $X\,^1\Sigma^+$ state of LiH provides important opportunities for comparison with both theoretical and experimental work, including the SCF[27] and CI[29] results of Wolfsberg and co-workers, and the work of Bishop and Cheung,[30] and Stwalley, Vidal, and co-workers discussed above. The essential aspect of the discrepancy between the theoretically and experimentally deduced forms for the AC for the $X\,^1\Sigma^+$ state of LiH will be resolved. The outline of the manuscript is as follows. Section II introduces the space fixed, center-of-mass fixed, and body fixed frame coordinate systems. The evaluation of $H(J,I,\mathbf{R})$ is considered and several relations are developed which provide extremely stringent tests of the algorithms used to evaluate $\mathsf{k}(J,I,\mathbf{R})$. Section III reports and analyzes the results of our calculations on the benchmark systems which are studied using extended Gaussian basis sets and second order CI wave functions. Section IV summarizes and concludes.

II. THEORETICAL APPROACH

In this work three coordinate systems are considered: (i) SFF coordinates **r**, **R**; (ii) CMFF coordinates **q**, **Q**, **C**; and (iii) BFF coordinates **w**, **W**, **B**, **C**. Here **r**, **q**, **w** denote the $3N^e$ coordinates of the N^e electrons in the SFF, CMFF, and BFF, respectively, **R** denotes the $3N$ SFF coordinates of the N nuclei, **Q** denotes $3N-3$ nuclear coordinates exclusive of translation, **W** denotes $3N-6$ (or $3N-5$) internal coordinates, **C** denotes the coordinates of the center of mass of the *entire* system, and **B** denotes the orientation of the molecule (body fixed) axis. The interrelation of these three coordinate

systems is well known.[5,22–25] For diatomic systems, which are to be considered in this work and for which the present approach obtains its maximal advantage, we collect the relevant results below. We have

$$\mathbf{C}=\left(\sum_i m_e\mathbf{r}^i+M_1\mathbf{R}^1+M_2\mathbf{R}^2\right)\Big/M^T, \tag{2.1}$$

$$\mathbf{q}^i=\mathbf{r}^i-\mathbf{O}, \tag{2.2a}$$

$$\mathbf{O}=(M_1\mathbf{R}^1+M_2\mathbf{R}^2)/M, \tag{2.2b}$$

$$\mathbf{Q}=\mathbf{R}^2-\mathbf{R}^1, \tag{2.3}$$

$$\begin{pmatrix} w_x^i \\ w_y^i \\ w_z^i \end{pmatrix}=\begin{pmatrix} -\sin\phi & \cos\phi & 0 \\ -\cos\theta\cos\phi & -\cos\theta\sin\phi & \sin\theta \\ \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \end{pmatrix}\begin{pmatrix} q_x^i \\ q_y^i \\ q_z^i \end{pmatrix}, \tag{2.4}$$

$$W=|\mathbf{R}^2-\mathbf{R}^1|, \tag{2.5}$$

$\mathbf{B}=(\theta,\phi)$ where θ, ϕ are the spherical polar coordinates of $\mathbf{Q}$ and the choice of axis system in Eq. (2.4) follows Kronig.[22] Here $M^T=M_1+M_2+m_eN^e$ is the total system mass and $M=M_1+M_2$ is the total nuclear mass. The total kinetic energy operator in the SFF, CMFF, and BFF is, respectively,

$$\hat{T}^{\mathrm{SFF}}=\hat{T}_e^r-\sum_K\frac{\Delta_K^R}{2M_K}, \tag{2.6a}$$

$$\hat{T}^{\mathrm{CMFF}}=\hat{T}_e^q+\hat{T}_C+\frac{\hat{\mathbf{P}}^q\cdot\hat{\mathbf{P}}^q}{2M}-\frac{1}{2\mu}\Delta^Q, \tag{2.6b}$$

$$\hat{T}^{\mathrm{BFF}}=\hat{T}_e^w+\hat{T}_C+\frac{\hat{\mathbf{P}}^w\cdot\hat{\mathbf{P}}^w}{2M}+\hat{T}_W+\hat{T}_{\mathrm{ROT}}, \tag{2.6c}$$

where

$$\hat{T}_W=\frac{-1}{2\mu}\frac{1}{W}\frac{\partial^2}{\partial W^2}W, \tag{2.7a}$$

$$\hat{T}_{\mathrm{ROT}}=\frac{-1}{2\mu W^2}\left\{\frac{1}{\sin\theta}\left(\frac{\partial}{\partial\theta}-i\hat{L}_x^w\right)\sin\theta\left(\frac{\partial}{\partial\theta}-i\hat{L}_x^w\right)+\frac{1}{\sin^2\theta}\left[\frac{\partial}{\partial\phi}-i(\cos\theta\,\hat{L}_z^w+\sin\theta\,\hat{L}_y^w)\right]^2\right\}, \tag{2.7b}$$

$$\hat{\mathbf{P}}^q=-i\sum_i\nabla_i^q, \tag{2.8}$$

$$\hat{\mathbf{L}}^w=-i\sum_i\mathbf{w}_i\times\nabla_i^w, \tag{2.9}$$

$$\hat{T}_C=\frac{-1}{2M^T}\Delta^C, \tag{2.10}$$

$$\hat{\mathbf{T}}_e^r=\frac{-1}{2}\sum_i\Delta_i^r \tag{2.11}$$

with $\mu^{-1}=M_1^{-1}+M_2^{-1}$ and $\Delta\equiv\nabla\cdot\nabla$. In the above atomic units have been used and when specifying an operator the following convention has been used. A superscript variable name is used to indicate the variable name used in the formal operator while the subscript index denotes the corresponding particle index, e.g.,

$$\nabla_i^q\equiv\left(\frac{\partial}{\partial q_x^i},\frac{\partial}{\partial q_y^i},\frac{\partial}{\partial q_z^i}\right).$$

The P^2 term in Eqs. (2.6b) and (2.6c) is referred to as the mass polarization term.

The electronic Schrödinger equation $(T_e+V-E^I)\psi_I=0$ is similar in the three coordinate systems. The difference is in the dependence of ψ_I on the nuclear coordinates. In the SFF $\psi_I\equiv\psi_I(\mathbf{r};\mathbf{R})$. In the CMFF $\psi_I\equiv\psi_I(\mathbf{q};\mathbf{Q})$. In the BFF $\psi_I\equiv\psi_I(\mathbf{w};\mathbf{W})$. Thus, the electronic wave function depends parametrically on six nuclear coordinates in the SSF, three nuclear coordinates in the CMFF, and only in the BFF does the electronic wave function depend parametrically on a single coordinate, the internuclear distance W.

From these observations several valuable equalities emerge. From the inverse of Eqs. (2.1)–(2.3) and the chain rule we have

$$M^2\Delta^Q=M_2^2\Delta_1^R+M_1^2\Delta_2^R-2M_1M_2\nabla_1^R\cdot\nabla_2^R \tag{2.12}$$

in the space spanned by the ψ_J. Then from Eqs. (2.6a) and (2.6b) the operator equivalence

$$\sum_{ij}\nabla_i^q\cdot\nabla_j^q=\Delta_1^R+\Delta_2^R+2\nabla_1^R\cdot\nabla_2^R \tag{2.13}$$

obtains so that

$$\mathrm{MP}(J,I,\mathbf{R})=\sum_{j\alpha}k_{j\alpha,j\alpha}(J,I,\mathbf{R})+\sum_\alpha\left[k_{1\alpha,2\alpha}(J,I,\mathbf{R})+k_{2\alpha,1\alpha}(J,I,\mathbf{R})\right], \tag{2.14}$$

where $j=1,2$, $\alpha=x,y,z$, the index $j\alpha$ refers to the SFF variable R_α^j and

$$-\mathrm{MP}(J,I,\mathbf{R})=\left\langle\psi_J\left|\sum_{ij}\nabla_i^q\cdot\nabla_j^q\right|\psi_I\right\rangle_{\mathbf{q}}. \tag{2.15}$$

In deriving Eq. (2.14) the equality

$$h_{\alpha,\beta}(J,I,\mathbf{R})=\frac{\partial}{\partial R_\alpha}g_\beta(J,I,\mathbf{R})-k_{\alpha,\beta}(J,I,\mathbf{R}) \tag{2.16}$$

and the antisymmetry of g_β, $g_\beta(J,I,\mathbf{R})=-g_\beta(I,J,\mathbf{R})$ which is valid for real-valued ψ_I have been used.

Equation (2.14) provides a stringent test of the methodology used in this work. The evaluation of $k_{\alpha,\beta}(J,I,\mathbf{R})$ requires the complete derivative apparatus discussed in Refs. 6–9, including the solution of the equations which give the derivatives of the molecular orbitals, the coupled perturbed state averaged MCSCF equations,[6] and the derivatives of the CI coefficients, the coupled perturbed CI (CPCI) equations.[8,9] The matrix element $\mathrm{MP}(J,I,\mathbf{R})$, on the other hand, is independent of the derivative apparatus. Thus, the verification of Eq. (2.14) provides a stringent test of the algorithms involved.

Another useful diagnostic relation can be obtained by comparing the CMFF and BFF expressions for $\hat{T}$ with ψ_J restricted to be $^1\Sigma^+$ states. In that subspace we have the operator equivalence

$$\frac{\hat{\mathbf{L}}^w\cdot\hat{\mathbf{L}}^w}{W^2}+\frac{1}{W}\frac{\partial^2}{\partial W^2}W=\Delta^Q \tag{2.17}$$

so that

$$\left(\frac{M}{W}\right)^2 L_y^2(J,I,\mathbf{R}) = \sum_j M_{j'}^2 k_{jx,jx}(J,I,\mathbf{R}) - M_1 M_2 [k_{1x,2x}(J,I,\mathbf{R}) + k_{2x,1x}(J,I,\mathbf{R})], \quad (2.18)$$

where $j = 1,2$, $j' = 3-j$,

$$L_y^2(J,I,\mathbf{R}) = \langle \psi_J | \hat{L}_y^{w2} | \psi_I \rangle, \quad (2.19)$$

and $\hat{L}^w$ is evaluated with origin at the center of mass of the molecule [**O** in Eq. (2.2b)]. As above, the value of Eq. (2.18) as a diagnostic obtains from its relating a quantity independent of nuclear displacement derivatives to quantities which depend explicitly on such derivatives.

The physical content of Eqs. (2.14) and (2.18) can be seen as follows. Let $\hat{\mathbf{t}}^N$ denote the operator which generates a translation of all the nuclei along the space fixed axes and $\hat{\rho}_y^N$ the operator which generates a rotation of the nuclei about the body fixed y axis located at the nuclear center of mass, i.e., $\hat{\mathbf{t}}^N = \Sigma_{K=1}^2 \nabla_K^R$ and

$$\hat{\rho}_y^N = \sum_{K=1}^{2} (R_z^K - O_z) \frac{\partial}{\partial R_x^K}.$$

Then Eqs. (2.14) and (2.18) can be reexpressed in terms of $\hat{\mathbf{t}}^N$ and $\hat{\rho}_y^N$ as

$$\mathrm{MP}(J,I,\mathbf{R}) = \sum_\alpha \langle \hat{t}_\alpha^N \psi_J | \hat{t}_\alpha^N \psi_I \rangle_\mathbf{r}, \quad (2.14')$$

$$L_y^2(J,I,\mathbf{R}) = \langle \hat{\rho}_y^N \psi_J | \hat{\rho}_y^N \psi_I \rangle_\mathbf{r}, \quad (2.18')$$

where $\alpha = x,y,z$. Thus Eqs. (2.14) and (2.18) are seen to relate derivatives of Born–Oppenheimer electronic wave functions with respect to noninternal nuclear degrees of freedom to matrix elements of electronic operators.

In Sec. III we will require the AC for a $^1\Sigma^+$ state. From Eq. (2.6c) this is given by

$$H(I,I,\mathbf{R}) = \frac{\mathrm{MP}(I,I,\mathbf{R})}{2M} + \frac{L^2(I,I,\mathbf{R})}{2\mu W^2} + \frac{K_R(I,I,\mathbf{R})}{2M}, \quad (2.20a)$$

where

$$K_R(I,I,\mathbf{R}) = \left[\frac{M_2}{M_1} k_{1z,1z}(I,I,\mathbf{R}) + \frac{M_1}{M_2} k_{2z,2z}(I,I,\mathbf{R}) - 2k_{1z,2z}(I,I,\mathbf{R})\right] \quad (2.20b)$$

and

$$L^2(I,I,\mathbf{R}) = L_x^2(I,I,\mathbf{R}) + L_y^2(I,I,\mathbf{R}) + L_z^2(I,I,\mathbf{R}).$$

In Eq. (2.20) derivatives with respect to only two space fixed coordinates, R_z^1, R_z^2, are required. Each derivative preserves the original molecular symmetry axis. This should be compared with the direct evaluation of Eq. (1.4). There derivatives with respect to six nuclear coordinates are required of which four (two unique) break the original molecular symmetry axis. Thus if Eq. (1.4) is used directly a lower point group (C_s), involving a larger, more costly, configuration state function (CSF) space is required to characterize the CI wave function. This problem is exacerbated by the CPCI equations, the most costly step in the derivative procedure, which must be solved for four degrees of freedom in this larger CSF space instead of the two degrees of freedom in the smaller CSF space required if Eq. (2.20) is used.

Similar economies obtain for triatomic systems. In this case of the nine nuclear displacements required to evaluate Eq. (1.4) directly in the SFF only the six which preserve the original plane of symmetry are required in the BFF treatment. Also C_s symmetry can be used in the characterization of the Born–Oppenheimer electronic wave functions. These computational efficiencies should facilitate the characterization of electronically nonadiabatic interactions in triatomic systems. This capability complements the recent theoretical work on the vibronic states of X_3 systems in the presence of a Jahn–Teller intersection by Mead, Truhlar, and coworkers.[10–12]

Note that the economies indicated above are obtained without any additional algorithm development beyond that required to determine the requisite one-electron integrals, provided the algorithms required for the evaluation of nuclear wave functions derivatives are already available. The requisite matrix elements, $\mathrm{MP}(J,I,\mathbf{R})$ defined in Eq. (2.15) and $L^2(J,I,\mathbf{R})$ defined in Eq. (2.19), are each of the form

$$A(J,I,\mathbf{R}) = \left\langle \psi_J(\mathbf{r};\mathbf{R}) \left| \sum_i \hat{\mathbf{a}}_i \cdot \sum_j \hat{\mathbf{a}}_j \right| \psi_I(\mathbf{r};\mathbf{R}) \right\rangle_\mathbf{r}, \quad (2.21a)$$

$$= \left\langle \psi_J(\mathbf{r};\mathbf{R}) \left| \sum_i \hat{\mathbf{a}}_i^2 + \sum_{i\neq j} \hat{\mathbf{a}}_i \cdot \hat{\mathbf{a}}_j \right| \psi_I(\mathbf{r};\mathbf{R}) \right\rangle_\mathbf{r}, \quad (2.21b)$$

where $\hat{\mathbf{a}}_i$ is a vector of anti-Hermitian operators. Thus,

$$A(J,I,\mathbf{R}) = \sum_{ij} \gamma_{ij}^{JI} \langle \phi_i | \hat{a}^2 | \phi_j \rangle + 2 \sum_{ijkl} \Gamma_{ijkl}^{JI} \langle \phi_i | \hat{\mathbf{a}} | \phi_j \rangle \cdot \langle \phi_k | \hat{\mathbf{a}} | \phi_I \rangle, \quad (2.22)$$

where $\boldsymbol{\gamma}^{JI}$ and $\boldsymbol{\Gamma}^{JI}$ are the square one- and two-particle transition density matrices required for the evaluation of $g_\beta(J,I,\mathbf{R})$[6,7] and $k_{\alpha,\beta}(J,I,\mathbf{R})$[8,9] and $\boldsymbol{\phi}$ are the orthonormal molecular orbitals. These density matrices are normalized such that

$$E_I(\mathbf{R}) = \sum_{ij} \gamma_{ij}^{JI} h_{ij} + \sum_{ijkl} \Gamma_{ijkl}^{JI} g(ij|kl), \quad (2.23)$$

where h_{ij} and $g(ij|kl)$ are the standard one- and two-electron integrals.

Finally note that in Eq. (2.20) the values of $k_{\alpha,\beta}(I,I,\mathbf{R})$ and $\mathrm{MP}(I,I,\mathbf{R})$ are independent of nuclear masses while the value of $L^2(I,I,\mathbf{R})$ is mass dependent through the definition of **O** [Eq. (2.2b)]. Hence $L^2(I,I,\mathbf{R})$ must be reevaluated for distinct mass combinations. This, however, is not a computationally significant limitation since the effort required to evaluate $L^2(I,I,\mathbf{R})$ is minor compared to that required to evaluate $\mathsf{k}(I,I,\mathbf{R})$. Additional simplifications in this regard are discussed in Sec. III [Eqs. (3.1) and (3.2)].

III. RESULTS AND DISCUSSION

The ground $X\,^1\Sigma^+$, and first excited $A\,^1\Sigma^+$, state of the BeH^+ and LiH benchmark systems are characterized by the principal electron configurations $1\sigma^2 2\sigma^2$ and $1\sigma^2 2\sigma 3\sigma$, respectively. The molecular orbitals used in this work were

developed from a complete active space state averaged MCSCF (SA-MCSCF)[36] procedure which includes all $^1\Sigma^+$ CSFs resulting from the distribution of four electrons in the 1σ–4σ orbitals. Two $^1\Sigma^+$ states were considered in the state averaging procedure with weight vector $\mathbf{w} = (0.55,0.45)$. This orbital space describes the above indicated principal electron configurations and also includes a correlating function for the 2σ orbital (the M–H σ bond, M = Li or Be) which serves to help balance the description of the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ reference spaces. The CI wave functions are second order wave functions, denoted as SOCI wave functions, relative to this active space. Although we will emphasize the $X\,^1\Sigma^+$ state in this study, this manifestly two state description is employed here with an eye toward the future work to be noted in Sec. IV.

Extended Gaussian basis sets were used. The hydrogen basis is the $(9s|4s)$ Gaussian basis of Siegbahn and Liu[37] augmented with two p polarization functions ($\alpha = 1.4,0.25$). The beryllium basis is the $(12s5p|7s3p)$ basis of Dykstra *et al.*[38] augmented with two d functions ($\alpha = 1.0,0.14$).[39] The lithium basis is the $(11s7p|5s4p)$ basis of McLean and Chandler[40] augmented with two d polarization functions ($\alpha = 0.4,0.1$). Using these basis sets the SOCI expansions contain 3856 CSFs and 3798 CSFs in C_{2v} symmetry for BeH^+ and LiH, respectively.

The values[42] of atomic masses used in this study are $M(^1H) = 1.007\,8250$, $M(^2H) = 2.014\,1018$, $M(^6Li) = 6.015\,1232$, $M(^7Li) = 7.016\,0045$, and $M(^9Be) = 9.012\,1825$, $m_e = 5.485\,803 \times 10^{-4}$ all in atomic mass units (amu) and we use $a_0 = 0.529\,177\,06$ Å, hartree = 219 474.64 cm^{-1} and will use $W = R(M–H) = R$.

The results of the calculations are summarized in Tables I–IV. Tables I and II present the AC for the $X\,^1\Sigma^+$ state of LiH and BeH^+, respectively, and compare our results with those of Bishop and Cheung (BC),[30] (BC1),[34] and Bardo, Kleinman, Raczkowski, and Wolfsberg (BKRW).[29] The specialized CI wave function of BC yields the lowest variational energies reported to date for LiH. It can be seen from Table I that at the near equilibrium geometries $R(LiH) = 2.50$ and $3.00\,a_0$ ($R_e \cong 3.015\,a_0$)[35] our results are in excellent accord with those of BC although our Born–Oppenheimer energy [E(BO)] is not nearly as low as that of BC. Agreement with the limited CI results of BKRW [at $R(LiH) = 3.015\,a_0$] is less satisfactory which appears to be due to the limitations of the BKRW treatment. For BeH^+ larger discrepancies are observed between our results and the SCF results of Kleinman and Wolfsberg.[27] Agreement improves considerably when our results are compared with the CI results of BC1. From these comparisons two conclusions appear justified. The SA-MCSCF/CI description espoused here provides a viable starting point for the description of non-Born–Oppenheimer effects. Further, adequate treatment of correlation effects is required for a reliable evaluation of the AC. This later observation is in accord with the conclusions of BKRW.

It was pointed out in the introduction that for $R > R_e$ the AC determined by BC differs qualitatively from that inferred by Chen, Harding, Stwalley, and Vidal (CHSV)[16] from a detailed analysis of the $A \rightarrow X$ emission in LiH. In

TABLE I. Born–Oppenheimer energy [E(BO)][a] and adiabatic correction [U][b] for the $X\,^1\Sigma^+$ state of LiH.

R[a]	E(BO)	$U(^7Li^1H)$	$U(^7Li^2H)$	$U(^6Li^1H)$	$U(^6Li^2H)$
2.500	– 8.033 8644	208.28	169.37	229.98	191.07
	– 8.052 43[c]	208.4[c]	170.6[c]	230.5[c]	192.7[c]
3.000	– 8.046 6328	201.16	165.30	222.69	186.82
	– 8.065 53[c]	201.7[c]	166.7[c]	223.6[c]	188.6[c]
3.015	– 8.046 6533	201.00	165.20	222.52	186.73
	– 8.034 7[d]	198.34[d]	163.82[d]		
3.030	– 8.046 6589	200.84	165.11	222.36	186.63
3.500	– 8.041 2221	196.84	162.87	218.28	184.31
	– 8.060 19[c]	197.6[c]	164.4[c]	219.4[c]	186.2[c]
4.500	– 8.015 1189	192.46	160.53	213.85	181.92
	– 8.033 81[c]			215.3[c]	
5.000	– 8.001 6425	191.55	160.13	212.96	181.54
	– 8.020 02			217.5	
5.500	– 7.989 6723	191.30	160.15	212.76	181.61
	– 8.010 06[c]	201.8[c]	167.4[c]	223.9[c]	189.5[c]
7.500	– 7.962 8294	191.61	161.01	213.31	182.71

[a] In atomic units.
[b] In cm^{-1}.
[c] Reference 30.
[d] Reference 29.

Table I it is seen that for $R > R_e$ the present calculations also differ from those of BC. Whereas the AC of BC rises sharply for $R > 4.5\,a_0$ the AC determined here continues to decrease, exhibiting a shallow minimum for large R. This dependence of the AC is in good accord with the results CHSV (their Fig. 2). To illustrate this point more quantitatively it is convenient to reexpress the AC in terms of mass independent[27,30] quantities U^{Li} and U^H defined as follows:

$$H(X\,^1\Sigma^+,X\,^1\Sigma^+,R) \equiv U(^xLi^yH) = U^{Li}/M(^xLi) + U^H/M(^yH), \tag{3.1}$$

where the mass dependence of $H(I,I,R)$ is made explicit in $U(^xLi^yH)$ and $x = 6,7$ and $y = 1,2$. Then

TABLE II. Born–Oppenheimer energy [E(BO)][a] and adiabatic correction [U][b] for the $X\,^1\Sigma^+$ state of BeH^+.

R[a]	E(BO)	$U(^9Be^1H)$	$U(^9Be^2H)$
2.4795	– 14.904 9068	255.09	222.89
	– 14.852 255[c]	245.74[c]	217.61[c]
	– 14.934 44[d]	258.1[d]	226.9[d]
2.4895	– 14.904 9090	254.83	222.76
2.4995	– 14.904 8951	254.59	222.64

[a] In atomic units.
[b] In cm^{-1}.
[c] Reference 27.
[d] Reference 34, $R = 2.48\,a_0$.

$$U^{H} = [U(^{7}Li^{1}H) - U(^{7}Li^{2}H)]/ [1/M(^{1}H) - 1/M(^{2}H)] \quad (3.2a)$$

$$= [U(^{6}Li^{1}H) - U(^{6}Li^{2}H)]/ [1/M(^{1}H) - 1/M(^{2}H)], \quad (3.2b)$$

and

$$U^{Li} = [U(^{6}Li^{1}H) - U(^{7}Li^{1}H)]/ [1/M(^{6}Li) - 1/M(^{7}Li)] \quad (3.3a)$$

$$= [U(^{6}Li^{2}H) - U(^{7}Li^{2}H)]/ [1/M(^{6}Li) - 1/M(^{7}Li)]. \quad (3.3b)$$

The results of this analysis are summarized in Table III which also compares the results of the present calculations with those of BC and CHSV. The results of CHSV are taken from their Table X using a spline interpolation to obtain the values of U^{H} at the requisite geometries. CHSW did not determine U^{Li} since, as will be discussed further below, U^{Li} has little influence on the observed spectra as a result of the comparatively large atomic mass of lithium and the small percentage change in $1/M(^{x}Li)$ for the isotopic substitution $^{7}Li \rightarrow {}^{6}Li$. The results in Table III demonstrate the semiquantitative agreement between the present determination of U^{H} and that of CHSV. The asymptotic values of U^{H} (U^{H}_{∞}) and U^{Li} (U^{Li}_{∞}) can be determined from accurate atomic calculations. Again satisfactory agreement between the present treatment and these more accurate values is obtained. In particular our calculations at $R = 15.0\ a_0$ yield $U^{H}_{\infty} = 60.2(60.2)$ cm$^{-1}\times$amu and $U^{Li}_{\infty} = 914.7(936.65)$ cm$^{-1}\times$amu which may be compared with the asymptotic values based on atomic calculations compiled in Ref. 30 which are given parenthetically.

Table IV decomposes the AC into three contributions from Eq. (2.20), a contribution from MP(I,I,R) (nuclear translation), a contribution from $L^{2}(I,I,R)$ (nuclear rotation), and a contribution from k(I,I,R) (internal motion). It is important to note that this decomposition is somewhat arbitrary, depending on the choice of coordinate systems. The decomposition obtained here is in fact distinct from that used by either BC or BKRW. For example, the three contributions to $H(I,I,R)$ from Eq. (2.20) are necessarily positive definite. This is not the case for the prescriptions of BKRW or BC. From this table it is seen that the MP contribution is by far the largest. This is related to the fact that the one-electron contribution to MP from Eq. (2.21b) is proportional to the expectation of the electronic kinetic energy operator. More significantly, for these systems the contribution attributable to internal nuclear motion derivatives, $K_R(I,I,R)$ in Table IV, exhibits only a limited geometry dependence when compared with the MP(I,I,R) and $L^{2}(I,I,R)$ contributions. While this cannot be the case generally,[9] in such favorable circumstances the geometry dependence of the AC could be deduced exclusively from the MP(I,I,R) and $L^{2}(I,I,R)$ terms. In these instances characterization of single state properties (see below) would not require the costly evaluation of wave function derivatives with respect to internal nuclear coordinates. A goal of future research will be the *a priori* determination of conditions for the validity of such an approximate treatment.

TABLE III. Analysis of the adiabatic correction: Decomposition into mass independent terms.

	U^{H}[a]			U^{Li}[a]	
R[b]	JY[c]	BC[d]	CHSW[e]	JY[c]	BC[d]
2.5	6.1	5.8	12.2	7.2	7.3
3.0	0.0	0.0	0.0	0.0	0.0
3.5	−3.8	−3.6	−5.99	−3.7	−3.8
4.5	−8.1	−7.2	−10.8	−5.8	−6.5
5.0	−9.0	−5.5	−12.0	−4.9	−3.8
5.5	−9.5	−1.0	−12.6	−2.9	8.2
7.5	−10.6		−13.1	7.0	

[a] In cm^{-1} amu.
[b] In a_0.
[c] This work relative to value at $R = 3.0$, $U^{H} = 72.4$, or $U^{Li} = 907.7$.
[d] Reference 30 relative to value at $R = 3.0$, $U^{H} = 70.5$, or $U^{Li} = 924.0$.
[e] Reference 16 relative to interpolated value at $R = 3.0$, $U^{H} = 0.10$.

TABLE IV. Analysis of the adiabatic correction[a] using Eq. (2.20).

	R[b]	MP[c]	$L2$[d]	KR[e]
$^{7}Li^{1}H$	2.5	124.53	58.89	24.86
	3.0	123.06	52.84	25.27
	3.5	122.04	49.63	25.17
	4.5	120.91	46.83	24.73
	5.0	120.64	46.21	24.69
	5.5	120.52	45.82	24.95
	7.5	121.05	45.38	25.18
$^{9}Be^{1}H^{+}$	2.4795	179.20	50.03	25.85
	2.4895	179.16	49.84	25.83
	2.4995	179.12	49.66	25.81

[a] In cm^{-1}.
[b] In a_0.
[c] MP$\equiv$MP(I,I,R)$/2M$ in Eq. (2.20).
[d] $L2 \equiv L^{2}(I,I,R)/2\mu R^{2}$ in Eq. (2.20).
[e] $KR \equiv K_R(I,I,R)/2M$ in Eq. (2.20).

The data in Tables I and II can be used to evaluate the equilibrium geometries for different isotopic combinations and therefore the shifts in equilibrium positions due to isotopic substitution. Using quadratic fits to this data we find for LiH, $\Delta R_e(^{7}Li^{1}H) = 3.85\times10^{-4}$ Å, $\Delta R_e(^{7}Li^{2}H) = 2.14\times10^{-4}$Å, $\Delta R_e(^{6}Li^{1}H) = 3.97\times10^{-4}$ Å, and $\Delta R_e(^{6}Li^{2}H) = 2.27\times10^{-4}$ Å. Here $\Delta R_e = R_e - R_e^{BO}$, R_e is the equilibrium geometry obtained from the PEC including the adiabatic correction, R_e^{BO} is the equilibrium geometry from the (mass independent) Born–Oppenheimer PEC and we have $R_e^{BO} = 1.602\,42$ Å. Thus, the shifts in R_e due to isotopic substitution are $\delta R_e(^{7}LiH) = 1.71\times10^{-4}(2.9\times10^{-4})$ Å, $\delta R_e(^{6}LiH) = 1.70\times10^{-4}(2.9\times10^{-4})$ Å, $\delta R_e(Li^{1}H) = 0.12\times10^{-4}(0.0\times10^{-4})$ Å, $\delta R_e(Li^{2}H) = 0.13\times10^{-4}(0.0\times10^{-4})$ Å. Here $\delta R_e(^{n}LiH) = \Delta R_e(^{n}Li^{1}H) - \Delta R_e(^{n}Li^{2}H)$, $\delta R_e(Li^{n}H) = \Delta R_e(^{6}Li^{n}H) - \Delta R_e(^{7}Li^{n}H)$ and the results from the experimental analysis of CHSV, which incorporates very accurate microwave data,[43] have been given parenthetically. The isotope shifts obtained here are similar to those reported by Kleinman and Wolfsberg[27] using SCF wave functions which from their Table I are $\Delta R_e(^{7}Li^{1}H) = 3.8\times10^{-4}$ Å, $\Delta R_e(^{7}Li^{2}H) = 2.2\times10^{-4}$ Å, and $\delta R_e(^{7}LiH) = 1.6\times10^{-4}$

Å. The present results are also in reasonable accord with those of CHSV. The direction of the isotope shift for hydrogen–deuterium substitution is correctly predicted although the magnitude is somewhat smaller than the observed values. Also note that the theoretical values of R_e are approximately 0.01 Å longer than the experimental values which is not unreasonable given the basis sets employed in this initial work.

A similar analysis applied to the BeH^+ data yields $\Delta R_e(^9Be^1H) = 3.67\times10^{-4}$ Å and $\Delta R_e(^9Be^2H) = 1.77\times10^{-4}$ Å, with $R_e^{BO} = 1.315\,48$ Å, so that $\delta R_e(^9BeH) = 1.9\times10^{-4}$ Å. This isotope shift is similar to that obtained above for LiH. The experimental values $R_e(^9Be^1H) = 1.3122$ Å[35] and $R_e(^9Be^2H^+) = 1.3113$ Å[43] yield $\delta R_e(^9BeH^+) = 9\times10^{-4}$ Å. In light of the preceding discussion our results indicate that the experimental value for $\delta R_e(^9BeH^+)$ is too large by at least a factor of 2. Note that unlike the more recent LiH analysis the experimental values for $R_e(^9Be^xH^+)$ were obtained from the observation of the $A\rightarrow X$ emission only.[43,44]

IV. SUMMARY AND CONCLUSIONS

The transformation to body fixed coordinates introduced by Kronig[22] for the study of spectra of diatomic molecules is used here to develop a computationally efficient algorithm for determining the total second derivative nonadiabatic coupling matrix element

$$H(J,I,\mathbf{R}) = \left\langle \psi_J(\mathbf{r};\mathbf{R}) \left| \sum_i \frac{-1}{2M_i} \frac{\partial^2}{\partial R_i^2} \psi_I(\mathbf{r};\mathbf{R}) \right\rangle_{\mathbf{r}} . \quad (4.1)$$

Here the Born–Oppenheimer electronic wave functions $\psi_J(\mathbf{r};\mathbf{R})$ are approximated by large scale CI wave functions developed from MCSCF reference spaces and orbitals. Only derivatives of the electronic wave function with respect to internal nuclear coordinates are required. These are evaluated using previously developed analytic derivative techniques. Identities [Eqs. (2.14) and (2.18)] are presented which provide stringent tests of the wave function derivative algorithms required for the evaluation of Eq. (4.1). These identities relate derivatives of the Born–Oppenheimer wave functions with respect to overall nuclear translation and rotation to matrix elements of the electronic operators P^2 (total electronic linear momentum) and L^2 (total electronic orbital angular momentum). The methodology introduced here will provide similar computational efficiencies for triatomic systems.

The adiabatic correction $[H(I,I,\mathbf{R})]$ to the Born–Oppenheimer potential energy curve for the $X\,^1\Sigma^+$ state of LiH is considered. Contracted Gaussian basis sets of better than double zeta polarization quality and second order configuration state function spaces are used to expand the Born–Oppenheimer wave functions. For the $X\,^1\Sigma^+$ state of LiH the adiabatic correction obtained here is in good accord with that inferred by Chen, Harding, Stwalley, and Vidal (CHSV)[16] from an analysis of LiH $A\rightarrow X$ emission, resolving the essential aspect of a discrepancy between the experimental and theoretical evaluations of this quantity. The results for the adiabatic correction were used to estimate shifts in equilibrium geometries (R_e) arising from isotopic substitution. Reasonable agreement between our theoretical values and those determined by CHSV (which incorporate accurate microwave data) was found for the hydrogen–deuterium induced isotope shift in R_e for the $X\,^1\Sigma^+$ state of LiH.

For the $X\,^1\Sigma^+$ states considered in this work the geometry dependence of the adiabatic correction (AC) is dominated by the P^2 and L^2 contributions to Eq. (2.20). In such favorable circumstances the present approach eliminates the need to evaluate wave function derivatives when treating the effects of isotopic substitution on single state properties such as R_e and ω_e. It will be a goal of future work to attempt to determine *a priori* criteria for the validity of such an approximate treatment.

The procedure developed in this work is applicable to both the interstate coupling matrix element $H(J,I,\mathbf{R})$, and the diagonal correction $H(I,I,\mathbf{R})$ emphasized here. The X, $A\,^1\Sigma^+$ states of LiH are known to undergo a charge transfer transition $Li^0H^0\rightarrow Li^+H^-$ near $R = 3.9$ Å.[15] This transition is reflected in the AC to the Born–Oppenheimer potential of the $A\,^1\Sigma^+$ state[15] but is not evident in the AC to the $X\,^1\Sigma^+$ state.[15,16] Consequently, it is relevant to consider the possibility of measurable nonadiabatic effects in the description of the $A\,^1\Sigma^+$ state. In a future work the methodology introduced here will be used to address this question using an adiabatic states analysis in which all interstate nonadiabatic couplings are explicitly included.

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